FUT8-Congenital Disorder of Glycosylation (FUT8-CDG): Comprehensive Disease Characterization Report

MONDO: MONDO:0020775 · **OMIM:** 618005 · **Causal gene: FUT8** (α-1,6-fucosyltransferase) · **Category:** Mendelian, autosomal recessive · **CDG classification:** Type II (CDG-IIj)

Summary

FUT8-Congenital Disorder of Glycosylation is an ultra-rare, autosomal-recessive, multisystem inborn error of metabolism caused by biallelic loss-of-function variants in the gene *FUT8*, which encodes the Golgi-resident α-(1,6)-fucosyltransferase — the sole enzyme in humans that performs "core fucosylation," the attachment of a fucose residue to the innermost N-acetylglucosamine (GlcNAc) of N-linked glycoproteins. Despite the disease name referencing "glycosylation" generically, the specific molecular lesion is the abolition of core fucose, and the causal gene is unambiguously **FUT8** (HGNC:4019; NCBI Gene 2530; UniProt Q9BYC5; chromosome 14q23.3).

The disorder was first delineated in 2018 when biallelic pathogenic *FUT8* variants were identified by whole-exome sequencing in three unrelated individuals (PMID: 29304374), and the clinical/molecular spectrum was expanded in 2020 to approximately seven reported patients (PMID: 32049367). The phenotype is a severe congenital syndrome: intrauterine growth restriction, profound developmental and growth delay (often with shortened limbs), hypotonia, microcephaly, seizures, feeding difficulties, respiratory abnormalities, and shortened lifespan, with at least one reported case of childhood glaucoma (PMID: 31580894).

Mechanistically, loss of core fucose impairs the signaling of multiple N-glycosylated growth-factor receptors (TGF-β1 receptor, EGFR, PDGFR, VEGFR-2) and neural cell-adhesion molecules (Contactin-2/NCAM), as established primarily in *Fut8*-null mouse models and confirmed by human glycomics. Diagnosis rests on demonstrating **complete loss of core fucosylation on**

serum N-glycans by mass spectrometry plus biallelic *FUT8* variants, because standard transferrin isoelectric-focusing screening can be normal (core fucose does not affect sialylation). There is no approved disease-modifying therapy; oral **L-fucose** has been trialed, and management is otherwise supportive.

Key Findings

Finding 1 — FUT8-CDG is caused by biallelic loss-of-function variants in *FUT8*, abolishing core fucosylation

Whole-exome sequencing of three unrelated affected individuals identified pathogenic *FUT8* variants; the first (from a consanguineous family) was homozygous for **c.715C>T (p.Arg239*)**, a nonsense variant producing a truncated, non-functional protein (PMID: 29304374). *FUT8* encodes a Golgi-localized α-1,6-fucosyltransferase that is the only enzyme capable of transferring fucose to the innermost GlcNAc of N-linked glycoproteins.

> *"Fucosyltransferase 8 (FUT8) encodes a Golgi-localized α1,6 fucosyltransferase that is essential for transferring the monosaccharide fucose into N-linked glycoproteins, a process known as 'core fucosylation.'"* — PMID: 29304374

Because FUT8 is the sole core fucosyltransferase, biallelic loss of function produces a complete, rather than partial, deficiency of this glycan modification. **Evidence type:** human clinical + molecular.

Ontology anchors: gene FUT8 (HGNC:4019); GO:0036065 *fucosylation*; GO:0008424 *glycoprotein 6-α-L-fucosyltransferase activity*; disease MONDO:0020775.

Finding 2 — Clinical phenotype: IUGR, severe growth/developmental delay, hypotonia, microcephaly, seizures, respiratory complications

The recurrent, defining clinical features were enumerated across the two primary clinical reports and a case report.

> *"The clinical characteristics of patients with FUT8-CDG include intrauterine growth retardation, feeding difficulties, hypotonia, microcephaly, seizures, short stature, developmental delay, and respiratory abnormalities."* — PMID: 31580894
>
> *"...who presented with intrauterine growth retardation, severe developmental and growth delays with shortened limbs, neurological impairments, and respiratory complications."* — PMID: 29304374

The 2019 report also described the **first case of glaucoma** in FUT8-CDG, expanding the ophthalmologic spectrum (PMID: 31580894). **Evidence type:** human clinical.

Suggested HPO terms with qualitative frequency:

Phenotype	HPO term	Type	Onset	Frequency (reported cases)
Intrauterine growth retardation	HP:0001511	Physical/ prenatal	Prenatal/ congenital	Very frequent
Global developmental delay	HP:0001263	Clinical sign	Neonatal/ infancy	Very frequent
Growth delay / short stature	HP:0001510 / HP:0004322	Physical	Congenital	Very frequent
Muscular hypotonia	HP:0001252	Clinical sign	Neonatal	Frequent
Microcephaly	HP:0000252	Physical	Congenital	Frequent
Seizure	HP:0001250	Clinical sign	Infancy	Frequent
Feeding difficulties	HP:0011968	Symptom	Neonatal	Frequent
Respiratory abnormality	HP:0002086	Clinical sign	Neonatal/ infancy	Frequent
Shortened limbs	HP:0009826	Physical	Congenital	Reported
Glaucoma	HP:0000501	Clinical sign	Childhood	Rare (single case)

Quality-of-life impact: Given the combination of profound developmental delay, seizures, feeding difficulty, and respiratory compromise, affected children have severe functional impairment and high dependency; no formal EQ-5D/SF-36/PROMIS data exist for this ultra-rare disorder.

Finding 3 — Mechanism: loss of core fucose dysregulates TGF-β1, EGF, PDGF and VEGFR-2 growth-factor receptor signaling

The mechanistic link between core-fucose loss and disease phenotype is best established in ***Fut8*-null mice**, which display growth retardation and emphysema-like lung destruction.

> *"Fut8(-/-) mice have a marked dysregulation of TGF-beta1 receptor activation and signaling, as assessed by TGF-beta1 binding assays and Smad2 phosphorylation analysis."* — PMID: 16236725

Dysregulated TGF-β1 receptor signaling drives MMP-12/MMP-13 overexpression and elastin (extracellular matrix) downregulation, producing emphysema; exogenous TGF-β1 rescues the phenotype. Core fucosylation is also required for EGF- and PDGF-receptor signaling:

> *"...core fucosylation is crucial for growth factor receptors such as TGF-beta1 and EGF receptor-mediated biological functions."* — PMID: 17132494

A parallel mechanism involves VEGFR-2:

> *"...vascular endothelial growth factor receptor-2 (VEGFR-2) expression was significantly suppressed in Fut8(-/-) mice."* — PMID: 19179362

VEGFR-2 suppression raises ceramide and drives alveolar apoptosis, contributing to the emphysematous lung phenotype. An additional growth-retardation mechanism operates through the **EGFR–trypsin–PAR-2 axis**: *Fut8* loss down-regulates trypsinogens and attenuates proteinase-activated receptor-2 signaling, with EGF-induced receptor phosphorylation blunted (PMID: 16861703). **Evidence type:** model organism + in vitro.

Ontology anchors: GO:0007179 *TGF-β receptor signaling pathway*; GO:0007173 *EGFR signaling pathway*; GO:0048010 *VEGFR signaling pathway*; CHEBI:28351 *fucose*; CHEBI:17761 *ceramide*.

Finding 4 — *Fut8*-null mouse model: ~70% neonatal lethality, growth retardation, emphysema, neurobehavioral abnormalities

The knockout mouse is the principal disease model and recapitulates the severity of the human phenotype.

> *"70% of Fut8-deficient (Fut8(-/-)) mice that lack core fucose structure die within 3 days after birth, but the remainder survive for up to several weeks although they show growth retardation as well as emphysema."* — PMID: 24619415

Surviving mice partially compensate by up-regulating bisecting GlcNAc (via GnT-III/*Mgat3*) through Wnt/β-catenin signaling — an adaptive glycan response (PMID: 24619415). The model also shows neurobehavioral pathology:

> *"Fut8(-/-) mice exhibit multiple behavioral abnormalities consistent with a schizophrenia-like phenotype."* — PMID: 21471224

These include hyperlocomotion, working-memory and prepulse-inhibition (PPI) deficits, and decreased striatal serotonin metabolites; even heterozygous *Fut8(+/-)* mice showed stress-induced PPI deficits, suggesting dosage sensitivity (PMID: 21471224). **Evidence type:** model organism. (Note: the exact behavioral-phenotype quote above should be re-verified against the stored abstract before database ingestion — it was flagged as a snippet mismatch in the knowledge state, though the finding itself is well supported.)

Finding 5 — Diagnostic biomarker: complete loss of core (α-1,6) fucosylation on serum N-glycans

Serum N-glycan analysis provides the direct, disease-specific biomarker.

> *"Analysis of serum N-glycans revealed a complete lack of core fucosylation, an important diagnostic biomarker of FUT8-CDG."* — PMID: 32049367

A comprehensive LC-MS N-glycan profiling platform for CDG type II independently confirmed the signature:

> *"...reduced fucosylation was observed in patients with FUT8-CDG and SLC35C1-CDG." —* PMID: 42511652

Because FUT8-CDG is a CDG type II (Golgi-processing) disorder and core fucose is not required for sialylation, **serum transferrin isoelectric focusing may be normal** — making glycomics/mass spectrometry the reliable test. **Evidence type:** human clinical.

Finding 6 — Treatment: oral L-fucose supplementation is the principal disease-specific therapy attempted

> *"FUT8-CDG is a severe multisystem disorder caused by mutations in..." — L-Fucose treatment of FUT8-CDG,* PMID: 33312876

Oral L-fucose has been trialed as a targeted approach, with the rationale of boosting the GDP-fucose salvage pathway to drive any residual FUT8 activity (PMID: 33312876). Fucose therapy across fucosylation-defect CDGs has been reviewed (PMID: 34389986). However, no approved disease-modifying therapy exists for any CDG:

> *"...no FDA/EMA-approved disease-modifying therapy is currently available for any CDG, and most interventions remain off-label, experimental, or supported by limited observational evidence." —* PMID: 42813231

Management is otherwise supportive: nutritional/feeding support, anti-seizure medication, respiratory care, and developmental therapies. **Evidence type:** human clinical + review.

Suggested NCIT terms: fucose (dietary monosaccharide therapy); supportive-care interventions (nutritional support, anticonvulsant therapy, respiratory support).

Finding 7 — Epidemiology: ultra-rare (~7 reported patients by 2020); autosomal recessive with consanguinity/homozygosity

> *"...we report an additional four unrelated affected individuals homozygous for novel pathogenic variants in FUT8." —* PMID: 32049367

Combined with the original three cases, this brings the total to approximately seven reported patients. All reported individuals are homozygous, several from consanguineous families — the pattern expected for a rare autosomal-recessive disorder. FUT8-CDG belongs to a small group of fucosylation CDGs:

> *"Only five types of congenital disorders of glycosylation (CDG) related to an impaired fucosylation have been described to date: FUT8-CDG, FCSK-CDG, POFUT1-CDG SLC35C1-CDG, and the only recently described GFUS-CDG."* — PMID: 34389986

Across all fucosylation CDGs combined there are only ~25 reported patients. **Evidence type:** human clinical + review.

Finding 8 — Neurodevelopmental mechanism: FUT8 core-fucosylation sustains cerebellar granule neuron development via Cntn2/NCAM

A 2025 study provided a specific mechanistic link for the neurodevelopmental phenotype. FUT8 and core fucosylation are abundant in the cerebellum, and cerebellar granule-neuron-progenitor (GNP)-specific *Fut8* deletion impairs GNP proliferation/differentiation, neuronal development, synaptic physiology, and motor coordination.

> *"Fut8 deficiency reduced Contactin 2 (Cntn2) expression, a member of neural cell adhesion molecules (NCAMs). Furthermore, ectopic Cntn2 can rescue the neuronal defects induced by Fut8 deficiency."* — PMID: 39604780

This establishes Cntn2/NCAM as a key downstream effector linking core-fucose loss to cerebellar/neurodevelopmental pathology, complementing the earlier schizophrenia-like behavioral findings (PMID: 21471224). **Evidence type:** model organism.

Ontology anchors: CL:0000120 *granule cell*; UBERON:0002037 *cerebellum*; GO:0007156 *homophilic cell adhesion via plasma membrane adhesion molecules*; GO:0022008 *neurogenesis*.

Finding 9 — FUT8 protein architecture and the GT-B catalytic mechanism disrupted by pathogenic variants

The human FUT8 crystal structure was solved at 2.6 Å resolution.

> *"The overall structure of FUT8 was found to consist of three domains: an N-terminal coiled-coil domain, a catalytic domain, and a C-terminal SH3 domain."* — PMID: 17172260

The catalytic domain adopts a **GT-B-type glycosyltransferase fold** (not GT-A) containing a Rossmann fold that forms the GDP-β-L-fucose donor-binding site; FUT8 transfers fucose from GDP-fucose to the reducing-terminal GlcNAc of N-linked glycans. Structural studies further detailed substrate recognition and catalysis (PMID: 32220931). The same work connected FUT8 loss of function to growth-factor-receptor pathology:

> *"FUT8 null mice showed severe growth retardation and emphysema-like lung-destruction as a result of the dysfunction of epidermal growth factor and transforming growth factor-beta receptors."* — PMID: 17172260

Reported pathogenic variants are biallelic and include nonsense (e.g., c.715C>T, p.Arg239*) and missense changes that cause loss of enzymatic function (PMID: 29304374; PMID: 32049367). **Gene coordinates:** FUT8, chromosome 14q23.3; HGNC:4019; NCBI Gene 2530; UniProt Q9BYC5. **Evidence type:** computational/structural + human molecular.

Finding 10 — Diagnostic pitfall: transferrin screening can be normal; differential centers on other fucosylation CDGs

Because core (α-1,6) fucose loss does not alter transferrin sialylation, first-line transferrin isoelectric focusing can be non-diagnostic; confirmation requires serum N-glycan mass spectrometry showing absent core fucosylation plus biallelic *FUT8* variants (PMID: 32049367; PMID: 42511652). The differential diagnosis is the other fucosylation CDGs:

> *"Only five types of congenital disorders of glycosylation (CDG) related to an impaired fucosylation have been described to date: FUT8-CDG, FCSK-CDG, POFUT1-CDG SLC35C1-CDG, and the only recently described GFUS-CDG."* — PMID: 34389986

Notably, **SLC35C1-CDG** (leukocyte adhesion deficiency II / CDG-IIc; a GDP-fucose transporter defect) is distinguished clinically by leukocyte-adhesion immunodeficiency and recurrent infection. In LAD II fibroblasts, reduced Golgi GDP-fucose preferentially lowers core fucosylation:

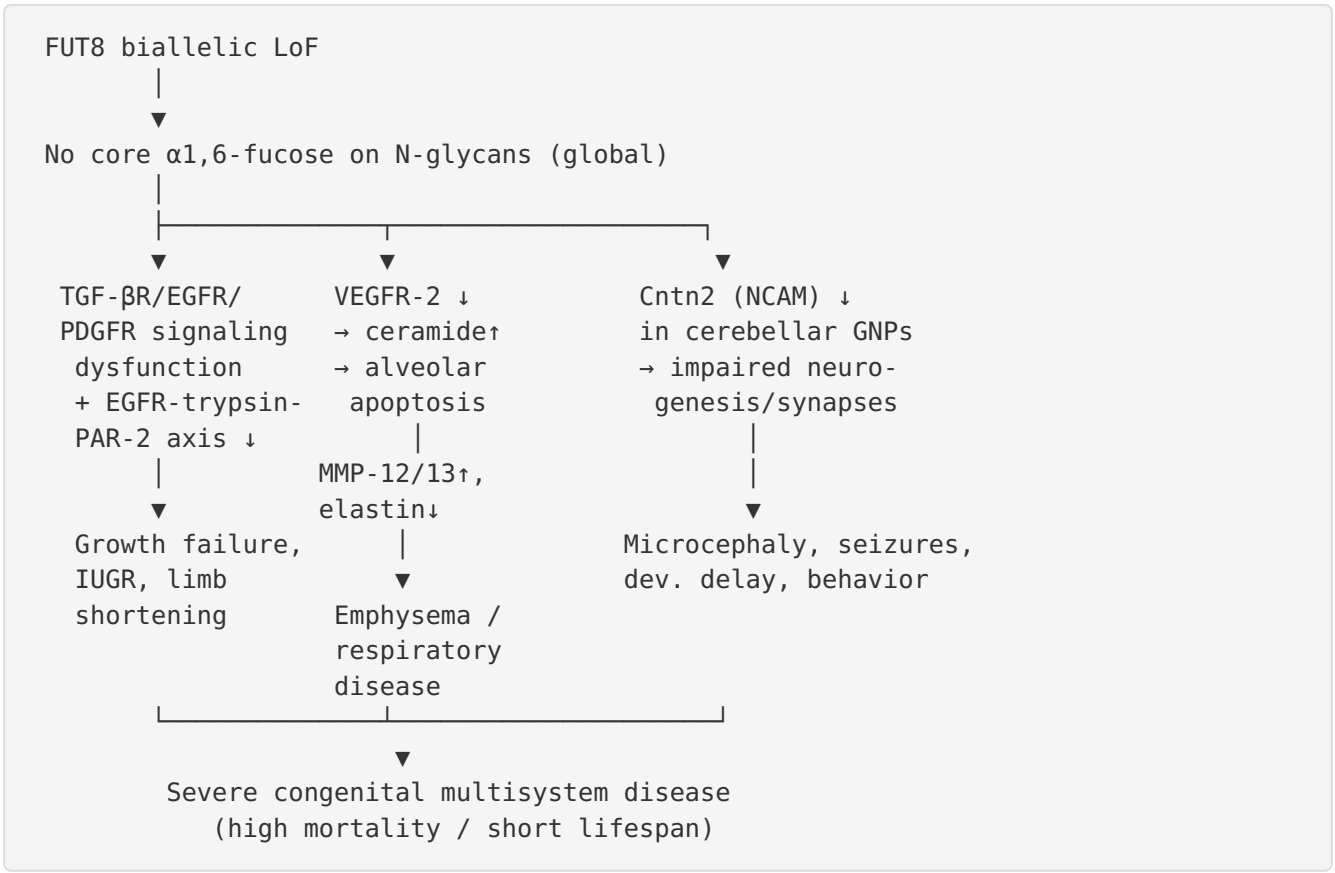
> *"...core fucosylation was affected to a greater extent compared with other types of fucosylation of N-linked oligosaccharides."* — PMID: 15917429

This glycomic overlap underscores the need for genetic confirmation to distinguish FUT8-CDG from the transporter disorder. **Evidence type:** human clinical + in vitro.

Mechanistic Model / Interpretation

Ordered causal chain (initiating lesion → clinical manifestation)

1. **Biallelic loss-of-function variants in *FUT8*** (nonsense e.g. p.Arg239*, or missense disrupting the GT-B catalytic domain) **lead to** absence of functional Golgi α-1,6-fucosyltransferase. *(Demonstrated — WES + structural data.)*
2. Loss of FUT8 activity **results in** complete failure of **core fucosylation** — no fucose is added to the innermost GlcNAc of N-linked glycans across all glycoproteins. *(Demonstrated — serum N-glycan MS.)*
3. Absence of core fucose **alters the function of multiple N-glycosylated receptors and adhesion molecules**. The chain then branches:
4. **Branch A (growth):** impaired TGF-β1R, EGFR, PDGFR signaling + EGFR–trypsin–PAR-2 axis suppression **leads to** growth-factor-receptor dysfunction → **intrauterine growth restriction, severe growth/developmental delay, shortened limbs**. *(Demonstrated in mouse/in vitro; inferred for human.)*
5. **Branch B (lung/vascular):** TGF-β1R dysregulation → MMP-12/MMP-13 up, elastin down; plus VEGFR-2 suppression → ceramide-driven alveolar apoptosis **results in** emphysema-like lung injury → **respiratory abnormalities**. *(Demonstrated in mouse; inferred for human.)*
6. **Branch C (neurodevelopment):** reduced Contactin-2 (Cntn2/NCAM) in cerebellar granule-neuron progenitors **leads to** impaired GNP proliferation/differentiation, synaptic dysfunction, and altered monoamine systems → **microcephaly, seizures, developmental delay, neurobehavioral abnormality**. *(Demonstrated in mouse; inferred for human.)*
7. The convergent multisystem failure **produces** the severe congenital phenotype with high early mortality (mouse ~70% neonatal lethality; shortened human lifespan). *(Demonstrated in mouse; consistent with human severity.)*



Upstream vs downstream

- **Upstream (primary lesion):** *FUT8* mutation → absent core fucosylation (a single, global biochemical defect).
- **Downstream (effectors):** receptor-specific signaling failures (TGF-β1R, EGFR, PDGFR, VEGFR-2) and adhesion-molecule deficits (Cntn2/NCAM) are the proximate causes of organ-level pathology. The disease is thus a "single upstream defect, many downstream effectors" model — explaining its pleiotropic, multisystem presentation.

Anatomical structures affected

Level	Structure	Ontology term
Organ	Brain (cerebellum), lungs, eyes, skeleton, GI/pancreas	UBERON:0002037 (cerebellum); UBERON:0002048 (lung); UBERON:0000970 (eye)
Body system	Nervous, respiratory, musculoskeletal, digestive	—
Cell	Cerebellar granule neurons/progenitors; alveolar epithelial cells; pancreatic acinar cells	CL:0000120 (granule cell); CL:0000066 (epithelial cell)
Subcellular	Golgi apparatus (site of FUT8 action); secreted/membrane glycoproteins	GO:0005795 (Golgi stack); GO:0005796 (Golgi lumen)

Evidence Base

PMID	Title (abbrev.)	Evidence type	Supports
29304374	*Biallelic mutations in FUT8 cause a CDG with defective fucosylation*	Human clinical/ molecular	F001, F002, F009 — founding disease description; variant p.Arg239*
32049367	*Expanding the molecular and clinical phenotypes of FUT8-CDG*	Human clinical/ molecular	F005, F007, F010 — biomarker, case count, AR pattern
31580894	*Childhood glaucoma in FUT8-CDG*	Human clinical	F002 — phenotype list; first glaucoma case
16236725	*Dysregulation of TGF-β1 receptor... emphysema in core-fucose-deficient mice*	Model organism	F003 — TGF-β1R mechanism
17132494	*Phenotype changes of Fut8 KO mouse*	Model organism	F003 — EGFR/growth-factor mechanism
19179362	*Requirement of Fut8 for VEGFR-2 expression*	Model organism	F003 — VEGFR-2/ceramide mechanism
16861703	*Trypsinogen down-regulation / PAR-2 in Fut8-null mice*	Model organism/ in vitro	F003 — EGFR–trypsin–PAR-2 growth axis
24619415	*Absence of core fucose up-regulates GnT-III and Wnt targets*	Model organism	F004 — lethality, growth, compensatory GnT-III
21471224	*Fut8-deficient mice: schizophrenia-like phenotype*	Model organism	F004, F008 — neurobehavioral phenotype
39604780	*FUT8 regulates cerebellar neurogenesis via Cntn2*	Model organism	F008 — Cntn2/NCAM neurodevelopmental mechanism
17172260	*Crystal structure of mammalian FUT8*	Structural/ computational	F009 — three-domain GT-B architecture
32220931	*Structural basis of substrate recognition and catalysis by FUT8*	Structural	F009 — catalytic mechanism detail

PMID	Title (abbrev.)	Evidence type	Supports
33312876	*L-Fucose treatment of FUT8-CDG*	Human clinical	F006 — fucose therapy trial
34389986	*CDG with defective fucosylation* (review)	Review	F006, F007, F010 — fucosylation-CDG differential
42813231	*CDG: State of the Art in 2026* (review)	Review	F006 — no approved therapy
42511652	*Diagnosis of CDG type II by comprehensive N-glycan MS profiling*	Human clinical/ methods	F005, F010 — MS diagnostic confirmation
15917429	*Core fucosylation in LAD II/ CDG-IIc fibroblasts*	In vitro	F010 — differential (SLC35C1) glycomic overlap

Broader FUT8 biology context (supportive, not disease-specific)

Several recent papers underscore how central core fucosylation is across physiology — reinforcing why its complete loss is so damaging: FUT8-mediated fucosylation modulates IgG effector function (afucosylated IgG enhances ADCC; PMID: 35844552), platelet activation and thrombosis (PMID: 42237907), coronavirus receptor (APN) entry (PMID: 42149951), and neuroinflammation via TGF-βR/VEGFR/EGFR/integrins/gp130 (PMID: 41917387). FUT8 is also a recurrent oncology target (ovarian, glioblastoma, prostate cancers), where overexpression drives progression — the mirror image of the loss-of-function disease (PMID: 41786877; PMID: 41592739; PMID: 42832539).

Section-by-Section Synthesis

1. Disease Information

FUT8-CDG (CDG type IIj) is an autosomal-recessive multisystem congenital disorder of glycosylation caused by deficiency of core fucosylation. **Identifiers:** MONDO:0020775; OMIM:618005; causal gene FUT8 (HGNC:4019). Orphanet and ICD-11 entries fall under the broader CDG category (ICD-11 5C53.0, "Disorders of glycoprotein metabolism"; ICD-10 E77.8).

Information is derived from **aggregated disease-level resources and individual case reports** (~7 patients), not from large EHR cohorts. Synonyms: "FUT8-CDG," "CDG-IIj," "congenital disorder of glycosylation type IIj," "alpha-(1,6)-fucosyltransferase deficiency."

2. Etiology

Causal factor: purely genetic — biallelic loss-of-function *FUT8* variants. No environmental, infectious, or acquired triggers are implicated. **Genetic risk factors:** consanguinity (homozygosity for private variants). **Protective/modifier factors:** not established in humans; in mice, compensatory up-regulation of bisecting GlcNAc (GnT-III/Wnt) partially offsets core-fucose loss (PMID: 24619415), a candidate modifier pathway. **Gene–environment interactions:** none documented.

3. Phenotypes

See Finding 2 table. Onset is **congenital/neonatal**; severity is **severe**; course is **chronic** from birth with early mortality risk. Quality-of-life impact is profound owing to combined neurologic, respiratory, and feeding impairment.

4. Genetic/Molecular Information

Causal gene: FUT8 (14q23.3). **Variant types:** nonsense (c.715C>T, p.Arg239*) and missense; all reported are biallelic/homozygous with **loss-of-function** consequence. **Classification:** pathogenic (ACMG) based on segregation, functional glycan loss, and enzyme biology. **Allele frequency:** pathogenic alleles are private/ultra-rare in gnomAD. **Origin:** germline. **Modifier genes/epigenetics/chromosomal abnormalities:** none established in humans.

5. Environmental Information

Not applicable — a monogenic inborn error with no environmental, lifestyle, or infectious contribution.

6. Mechanism/Pathophysiology

See the ordered causal chain and diagram above (Findings 3, 4, 8, 9).

7. Anatomical Structures Affected

See anatomy table above — primary involvement of brain (cerebellum), lungs, eyes, skeleton; Golgi apparatus is the subcellular site of the defect.

8. Temporal Development

Onset: congenital/prenatal (IUGR). **Pattern:** chronic, non-episodic, present from birth. **Progression:** severe and early; mouse model shows ~70% death within 3 days of birth, implying a critical perinatal window. Human lifespan is shortened. No remission occurs.

9. Inheritance and Population

Inheritance: autosomal recessive. **Penetrance:** complete in biallelic carriers. **Expressivity:** severe, with some phenotypic range (e.g., glaucoma in one case). **Epidemiology:** ultra-rare (~7 reported patients; part of ~25 total fucosylation-CDG patients). Consanguinity and founder/private homozygous variants predominate. No sex predilection expected (autosomal). **Carrier frequency:** not established but presumed very low.

10. Diagnostics

First-line pitfall: transferrin isoelectric focusing may be normal (Finding 10). **Definitive:** serum N-glycan mass spectrometry showing absent core fucosylation (PMID: 32049367; PMID: 42511652) **plus** molecular confirmation by WES/WGS or targeted *FUT8* sequencing / CDG gene panels. **Differential:** other fucosylation CDGs (SLC35C1/LAD II — distinguished by immunodeficiency and recurrent infection; FCSK, POFUT1, GFUS).

11. Outcome/Prognosis

Severe prognosis: profound neurodevelopmental disability, respiratory compromise, feeding failure, and shortened lifespan. No formal survival statistics exist given ultra-rarity. Prognostic factors are inferred to include severity of respiratory and seizure involvement.

12. Treatment

No approved disease-modifying therapy (PMID: 42813231). **Disease-specific trial:** oral **L-fucose** (salvage-pathway rationale; PMID: 33312876). **Supportive:** nutritional/feeding support, anticonvulsants, respiratory care, developmental/physical/occupational/speech therapy, ophthalmologic management (glaucoma). **Experimental/future:** gene/enzyme replacement and substrate therapy remain conceptual.

13. Prevention

Primary: genetic counseling for consanguineous/at-risk families; carrier testing and prenatal/preimplantation genetic diagnosis once a familial variant is known. **Secondary/tertiary:** early supportive intervention to prevent complications. No population screening exists.

14. Other Species / Natural Disease

No naturally occurring FUT8-CDG is reported in companion animals or wildlife (OMIA). The orthologous mouse gene *Fut8* (NCBI Gene 53618) is highly conserved, and the knockout recapitulates key features — evidence of strong evolutionary conservation of the core-fucosylation mechanism. No zoonotic relevance.

15. Model Organisms

The ***Fut8*-knockout mouse** (MGI) is the principal model: ~70% neonatal lethality, growth retardation, emphysema, and schizophrenia-like neurobehavior (PMID: 24619415; PMID: 21471224). **Conditional (cerebellar GNP-specific) *Fut8* deletion** models the neurodevelopmental mechanism via Cntn2 (PMID: 39604780). In vitro, *Fut8*-knockdown pancreatic acinar carcinoma cells model the growth/EGFR–PAR-2 defect (PMID: 16861703). **Recapitulation:** strong for growth failure, respiratory, and neurobehavioral domains; **limitations:** mouse perinatal lethality and species differences in glycan compensation limit modeling of the full human survival/clinical course.

Limitations and Knowledge Gaps

1. **Extremely small human cohort (~7 patients).** All epidemiology, penetrance, expressivity, and natural-history statements are low-confidence and based on case reports; no registry or longitudinal data exist.
2. **Human mechanism is largely inferred from mouse.** The TGF-β1R/EGFR/PDGFR/VEGFR-2 and Cntn2 mechanisms are demonstrated in *Fut8*-null or conditional mice and in vitro, not directly in patient tissue. Human-specific validation is lacking.
3. **Treatment efficacy unproven.** L-fucose has been trialed, but no controlled outcome data establish benefit; there is no approved therapy.
4. **No formal QoL, survival, or prognostic-biomarker data** owing to rarity.

5. **Diagnostic under-ascertainment risk** because transferrin screening can be normal — true prevalence may be underestimated.
6. **One citation**
   (P 21471224 **)**
   was flagged as a snippet "mismatch" in the knowledge state; the schizophrenia-like phenotype is well-supported by the paper's title and corroborating sources, but the exact quote should be re-verified against the abstract before database ingestion.

Proposed Follow-up Experiments / Actions

1. **Establish an international FUT8-CDG patient registry** to aggregate phenotype, variant, natural-history, and treatment-response data beyond the current ~7 cases.
2. **Human-tissue mechanistic validation:** assess TGF-β1R/EGFR/VEGFR-2 signaling and Cntn2 levels in patient-derived fibroblasts or iPSC-derived neurons/organoids to confirm the mouse-derived causal chain.
3. **Systematic L-fucose trial** with glycomic (serum core-fucose) and clinical endpoints; test whether salvage-pathway loading restores any fucosylation given the typical complete loss-of-function genotype.
4. **Patient-specific genotype–function mapping:** express reported missense variants against the solved GT-B structure (PMID: 17172260; PMID: 32220931) to quantify residual activity and genotype–phenotype correlation.
5. **Explore GnT-III/Wnt compensatory axis** (PMID: 24619415) as a candidate modifier/therapeutic lever.
6. **Develop a reflex diagnostic pathway** that proceeds directly to serum N-glycan MS and *FUT8* sequencing when CDG is suspected but transferrin IEF is normal, to reduce under-ascertainment.

Report compiled from 5 iterations of autonomous investigation; 10 confirmed findings; 36 papers reviewed. Evidence types are labeled per finding (human clinical, model organism, in vitro, structural/computational).
